

Steps for Installation of OpenMP

- **STEP 1: Check the GCC version of the compiler**

```
gcc --version
```

GCC provides support for OpenMP starting from its version 4.2.0. So if the system has GCC compiler with the version higher than 4.2.0, then it must have OpenMP features configured with it.

```
akbar@ubuntu: ~/Desktop
File Edit View Search Terminal Help
akbar@ubuntu:~/Desktop$ gcc --version
gcc (Ubuntu 7.3.0-27ubuntu1~18.04) 7.3.0
Copyright (C) 2017 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

akbar@ubuntu:~/Desktop$
```

If the system doesn't have the GCC compiler, we can use the following command

```
sudo apt install gcc
```

For more detailed support for installation, we can refer [here](#)

- **STEP 2: Configuring OpenMP**

We can check whether the OpenMP features are configured into our compiler or not, using the command

```
echo |cpp -fopenmp -dM |grep -i open
```

```
akbar@ubuntu: ~/Desktop
File Edit View Search Terminal Help
akbar@ubuntu:~/Desktop$ gcc --version
gcc (Ubuntu 7.3.0-27ubuntu1~18.04) 7.3.0
Copyright (C) 2017 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

akbar@ubuntu:~/Desktop$ echo |cpp -fopenmp -dM |grep -i open
#define _OPENMP 201511
akbar@ubuntu:~/Desktop$
```

If OpenMP is not featured in the compiler, we can configure it use using the

command

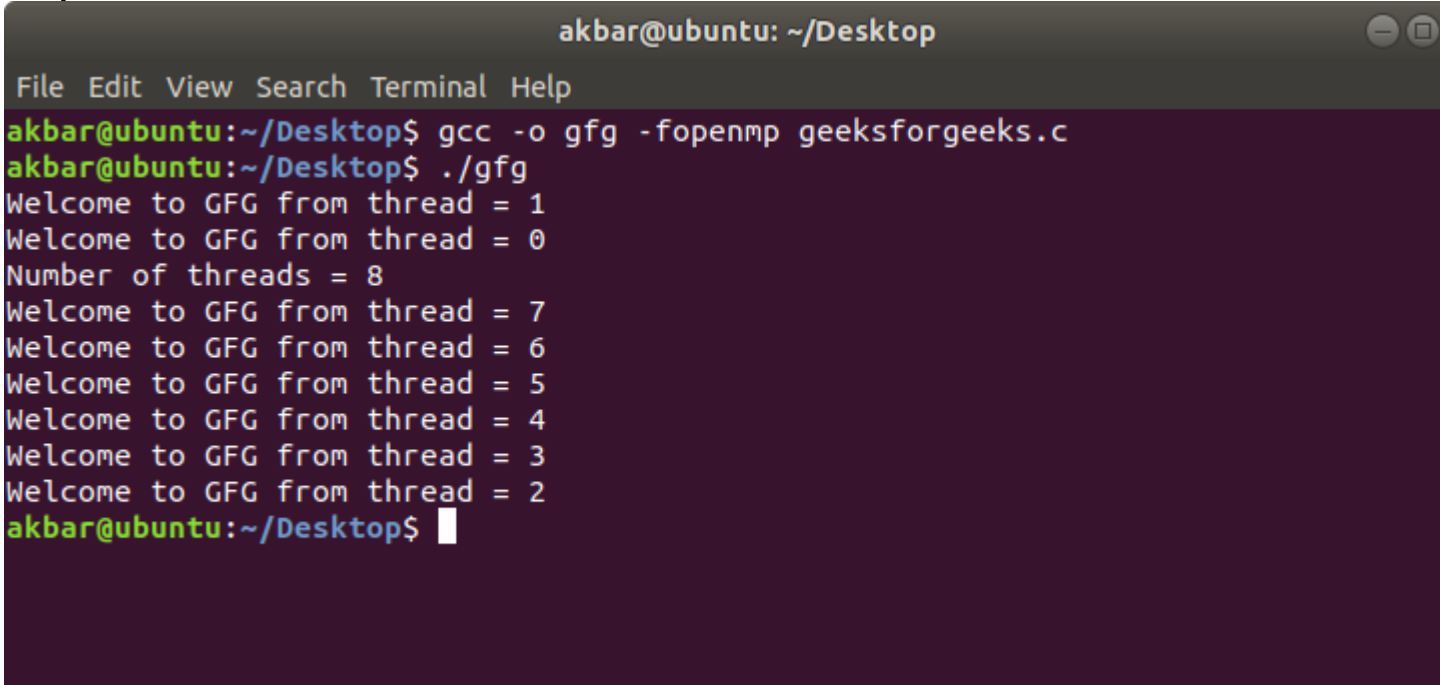
sudo apt install libomp-dev

- **STEP 3: Setting the number of threads**

In OpenMP, Before running the code, we can initialise the number of threads to be executed using the following command. Here, we set the number of threads to be getting executed to be 8 threads.

export OMP_NUM_THREADS=8

Output:

A terminal window titled 'akbar@ubuntu: ~/Desktop' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the following commands and output:

```
akbar@ubuntu:~/Desktop$ gcc -o gfg -fopenmp geeksforgeeks.c
akbar@ubuntu:~/Desktop$ ./gfg
Welcome to GFG from thread = 1
Welcome to GFG from thread = 0
Number of threads = 8
Welcome to GFG from thread = 7
Welcome to GFG from thread = 6
Welcome to GFG from thread = 5
Welcome to GFG from thread = 4
Welcome to GFG from thread = 3
Welcome to GFG from thread = 2
akbar@ubuntu:~/Desktop$
```

This program will print a message which will be getting executed by various threads.

Compile:

gcc -o gfg -fopenmp geeksforgeeks.c

Execute:

./gfg